Successive Cancellation Decoding of Reed-Solomon Codes

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Reference

[Trifonov(2014)] P. V. Trifonov.
Successive cancellation decoding of reed–solomon codes.
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Polar Subcodes

Consider the generator of the polar code $\mathbf{G}_p = \mathbf{B}_m \mathbf{F}_2^{\otimes m}$. Since $\mathbf{G}_p$ is invertable, any set of codeword $\mathbf{c} \in \mathcal{C}$ can be obtained with some constrained $\mathbf{u}$

$$\mathbf{c} = \mathbf{u}\mathbf{G}_p, \quad \mathbf{c}\mathbf{H}^T = \mathbf{u}\mathbf{G}_p\mathbf{H}^T = \mathbf{0}.$$

Consider $\mathbf{V} = \mathbf{E}\mathbf{H}\mathbf{G}_p^T$ as the reduced constraint matrix for some $\mathbf{E} \in \mathrm{GL}_{n-k}$, and denote the dynamic frozen bits as j_i for $i \in [0, n-k]$.

$$u_{j_i} = \sum_{s<j_i} V_{i,s} u_s.$$

The generator matrix of the corresponding code is therefore $\mathbf{G} = \mathbf{W}\mathbf{G}_p$ where $\mathbf{W}\mathbf{V}^T = \mathbf{0}$

RS code as a polar subcode

Now let us consider the extended Reed-Solomon codes over $\mathbb{F}_2^m$.

- The codewords are the polynomials

$$\mathbf{c} = (f(x_0), \ldots, f(x_{n-1})) \in V_n(\mathbb{F}_{2^m}), \quad f(x) = \sum_{j=0}^{k-1} f_j x^j \in \mathbb{F}_{2^m}[x].$$

- The expansion of the evaluation point $x = \sum_{i=0}^{m-1} x_i a_i$ in some basis $\{a_0, \ldots a_{m-1}\}$ of $\mathbb{F}_{2^m}$.
- The binary expansion of integer indices $j = \sum_{s=0}^{m-1} j_s 2^s$.

RS code as a polar subcode

Now, for $x \in \mathbb{F}_{2^m}$, we have the j-th power

$$x^j = \Big(\sum_{i=0}^{m-1} x_i a_i\Big)^{\sum_{s=0}^{m-1} j_s 2^s} = \prod_{s=0}^{m-1} \Big(\sum_{i=0}^{m-1} x_i a_i\Big)^{j_s 2^s} = \prod_{s=0}^{m-1} \Big(\sum_{i=0}^{m-1} x_i^{2^s} a_i^{2^s}\Big)^{j_s}$$

$$\Rightarrow x^j = \prod_{s=0}^{m-1} \Big(\sum_{i=0}^{m-1} x_i a_i^{2^s}\Big)^{j_s}.$$

Therefore x^j is a polynomial in variables $(x_0, \ldots, x_{m-1})$ of degree $\mathrm{wt}(j) = \sum_{s=0}^{m-1} j_s$, with nonzero degrees $Q_j = \{t = (t_0, \ldots, t_{m-1})\}$.

$$x^j = \sum_{t \in Q_j} w_{j,t} \prod_{i=1}^{m-1} x_i^{t_i}.$$

where the weights $w_{j,t}$ are calculated accordingly.

RS code as a polar subcode

And therefore the entries of codeword can be expressed as

$$f(x) = \sum_{j=0}^{k-1} f_j x^j = \sum_{j=0}^{k-1} f_j \sum_{t \in Q_j} w_{j,t} \prod_{i=0}^{m-1} x_i^{t_i}.$$

Observe that row $t = \sum_{j=0}^{m-1} t_{m-1-i} 2^i$ of $\mathbf{G}_p$ is the sequence of $\prod_{i=0}^{m-1} x_i^{1-t_i}$ evaluating at point $(x_0, \ldots, x_{m-1})$.

Thus, any codewords of the RS code can be written as

$$\mathbf{c} = \mathbf{f}\mathbf{W}\mathbf{G}_p,$$

with $\mathbf{f} = (f_0, \ldots, f_{k-1})$ the coefficient vector and $\mathbf{W} = (w_{j,t})$ the dynamic frozen matrix.

The resulted frozen set

Consider the bit-reversal function of an integer $i = (i_0, \ldots, i_{m-1})$ of degree m

$$r_m(i) = \sum_{j=0}^{m-1} i_j 2^{m-1-j}.$$

The resulted dynamic frozen set from the above transformation is

$$\mathcal{F} = [2^m] \backslash \{2^m - 1 - r_m(i) : 0 \leq i < k\},$$

Where we can see that the subchannels with index i having weight $\mathrm{wt}(i) < m - r$ is set to be frozen, with $r = \max_{0 \leq j < k} \mathrm{wt}(j)$.

This is proved by grouping the coefficients of the polynomial in the same cyclotomic coset C_s, and then verify the resulted linear independency of the columns by induction.

Invariant Permutation

Notice the resulted frozen set is independet of the choice of the basis of $\mathbb{F}_{2^m}$, while still induce different permutations of the decoded vectors. For a new basis $\mathcal{A}_i = (a_{i,0}, \ldots, a_{i,m-1})^T$, we constructs a new permuted sequence z_0^{n-1} from the originally received sequence y.

1. For each position, j $0 \leq j \leq n-1$, look at its binary representation $j = \sum_{s=0}^{m-1} j_s 2^s$.
2. Transform j into the new basis $\beta = \sum_{s=0}^{m-1} a_{i,s} j_s$.
3. Place the received symbol y_β, and place it in position j of the new sequence $z_j = y_\beta$.

Then the resulted new sequence z_0^{n-1} is still a valid RS noisy codeword, in a new basis $\mathcal{A}_i$, with the same frozen set.

Permutation Decoding

For a received signal y_0^{n-1},

1. Choose a basis $\mathcal{A}_i$ from the list.
2. Construct the permuted sequence z_0^{n-1} from last page.
3. Run SC decoding on this permuted sequence z.
4. Calculate the probability of the resulting decoded codeword.
5. Repeat for N different bases, and report the decoded codeword with the highest probability.

Extending SC to $\mathbb{F}_q$

To extend SC to $\mathbb{F}_{2^\mu}$ for some $\mu > 1$ ($m > \mu$ considering frozen bits), the conditional probability is modified to summing over all $u_i \in \mathbb{F}_{2^\mu}$,

$$W_n^{(2i)}(u_0^{2i}|y_0^{n-1}) = \sum_{u_{2i+1} \in \mathbb{F}_{2^\mu}} W_{\frac{n}{2}}^{(i)}(u_{0,ev}^{2i+1} + u_{0,od}^{2i+1}|y_0^{\frac{n}{2}-1}) W_{\frac{n}{2}}^{(i)}(u_{0,od}^{2i+1}|y_0^{n-1}).$$

Using fast Hadamard transform to compute this cyclic convolution is with complexity $O(\mu 2^\mu)$.

Furthermore, if transmitting a binary image of a code over BSC,

$$W_n^{(i)}(u_0^i|y_0^{n-1}) = P_{U_0^i|Y_0^{n-1}}(u_0^i|y_0^{n-1}) = \prod_{j=0}^{\mu-1} P_{U_{0,j}^i|Y_0^{n-1}}(u_{0,j}^i|y_0^{n-1}),$$

And therefore the decoding is performing, for information bits,

$$\hat{u}_i = \arg\max_{a \in \mathbb{F}_{2^\mu}} P_{U_0^{i-1}, Y_0^{n-1}|U_i}(\hat{u}_0^{i-1}, y_0^{n-1}|a)$$

Resulted frozen bits and channel reliability

The capacity of the i-th subchannel

$$P_{U_0^{i-1},Y_0^{n-1}|U_i}(\hat{u}_0^{i-1}, y_0^{n-1}|u_i)$$

converge to $\{0,\ldots,\mu\}$[15]. For subchannels with capacity close to $i \in \{1,\ldots,\mu-1\}$, the error probability is close to $1-2^{i-\mu}$.

Under BSC we have the decoding error probability bounded

$$p_i \leq Z_{n,i} = O(p^{2^{\mathrm{wt}(i)}}), \quad P \leq O(n\mu p^{2^{w_{\min}}}),$$

with p the Bhattacharyya parameter, and $w_{\min}$ the minimum weight of the information bit index. For our resulted frozen set,

$$w_{\min} \geq m - r$$

allows the decoding error probability to decrease sufficiently fast.